

Replace Cookie depositor controller

For Plant manager, Ops lead, QA
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Read time · 85 sec
Decision by 23 May 2026

Hypothesis · modelled, partial data

TECHNICAL PROPOSAL

Replace the Line 2 depositor PLC with an IPC + servo controller, hold dough weight tolerance under ±1.5 g.

Today the depositor uses a 2009-era PLC that drifts ±4 g during a shift, costing dough waste and triggering QA rejections. Modern IPC + closed-loop servo control holds tolerance, exposes line metrics to MES, and unblocks the new gluten-free SKU launch.

EFFORT

8 weeks

1 controls eng + 0.5 ops eng



LINE 2 / DEPOSITOR + CONVEYOR

WHAT CHANGES	
Depositor controller	Allen-Bradley PLC (2009) → Beckhoff IPC + EtherCAT servos
Dough weight tolerance	±4 g drift / shift → ±1.5 g closed-loop
Line metrics	paper logs → live to MES via OPC-UA
Recipe changeover	14 min manual setup → 3 min recipe recall

COST & VALUE		value
–	Hardware (IPC + servos + IO)	€48k /one-off E
–	Engineering effort	€22k /one-off E
+	Avoided dough waste	€31k /yr M
+	Faster changeovers (capacity)	€44k /yr E
+	Unblocks gluten-free SKU launch	€180k /yr G

TRADE-OFFS

Beckhoff IPC + EtherCAT (chosen) CHOSEN
+ Already in stock on Line 4, in-house expertise
– Vendor lock-in on TwinCAT

Siemens TIA Portal
+ Industry standard, broader spare-parts pool
– New vendor relationship, higher integration cost

Refurbish existing PLC
+ Lower upfront cost
– Does not solve drift; spare parts EOL in 2027

RISKS

Medium **Operational** · New HMI requires retraining — 2 day course for 6 operators E

Low **Performance** · Vendor reference site reports stable ±0.8 g over 6 months M

Medium **Complexity** · OPC-UA mapping touches MES schema — coordinate with IT E

Low **Tech debt** · Same controller family as Line 4, reduces overall sprawl M

risk surface

OPEN QUESTIONS FOR REVIEW

- Do we keep the old PLC in storage as a 1-week fallback or scrap immediately?
- Is QA willing to bank the gluten-free SKU launch on the new controller, or stays as gut?

- Who owns the OPC-UA tag schema — controls eng or MES team?
- Acceptable cap on auto-reject rate before alerting the line lead?

THE ASK

Approve build. Start sprint 22.

Reviewers: Plant manager · Ops lead · QA lead

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Behind Cookie depositor controller

MOTIVATION & PROBLEM

Line 2 depositor weight drift caused 6 QA holds in Q1 2026, scrapping ~2,400 kg of dough. The controller is also the bottleneck for the gluten-free SKU we have been asked to ship by Q4 — its recipe needs sub-2-gram weight precision that the current PLC cannot hold over a full shift. The PLC platform also goes end-of-life on parts support in 2027, so we will replace it within 18 months regardless.

DESIGN APPROACH

New controller is a Beckhoff CX5230 IPC running TwinCAT 3 with EtherCAT-connected servos on each depositor head. Closed-loop weight feedback from in-line checkweighers feeds the cycle controller every 50 ms. HMI is on the existing Line 2 panel; recipe storage moves to the plant MES. No mechanical changes — only the control cabinet, wiring, and HMI software change.

DEPENDENCIES

IT / MES team	OPC-UA broker provisioned · committed
QA	Validates new control limits before go-live
Vendor (Beckhoff)	Hardware lead time 4 weeks, slot reserved
Maintenance team	Owns spare-parts plan + commissioning sign-off

TESTING PLAN

Bench test in maintenance bay with the existing depositor head wired to the new IPC. Run 200 simulated recipe cycles, verify ±1.5 g over a 4-hour soak. On the line: dual-mode for 1 week (old PLC owns production, new IPC shadows), then cutover during scheduled Saturday window.

ROLLOUT PLAN

Saturday cutover with vendor on-site. Line 2 starts Monday on new controller for one SKU only (chocolate-chip), all others wait until QA signs off — typically 5 days. Old PLC stays in cabinet for 30 days as physical fallback before being decommissioned.

ROLLBACK

Pre-built switchover kit lets us revert to the old PLC in under 90 minutes by swapping cabinet wiring. Recipe data on the old PLC is preserved untouched until decommissioning.

MONITORING & METRICS

Dashboards: depositor weight p50/p95/p99 per SKU, auto-reject rate, recipe changeover time, controller uptime. Alarm on weight p99 outside ±2 g for >10 cycles, or auto-reject rate >2%.

REFERENCES

Line 4 commissioning report (Q3 2024) · vendor reference visit Wuppertal · ADR #017 (controller standardisation) · QA acceptance criteria doc rev 4.